



PROJECT:

Systems & Data Architecture for Real-Time Naval Fleet Monitoring (HMI)



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1. EXECUTIVE SUMMARY

This technical report proposes a high-performance, containerized Human-Machine Interface (HMI) designed to process, filter, and visualize massive maritime data streams. The core objective of this project was to transition from legacy, unindexed tabular file storage (CSV) containing more than **1.1 million Automatic Identification System (AIS)** records to an enterprise-level relational database management system (RDBMS) architecture.

By leveraging **Docker**, **PostgreSQL**, **SQLAlchemy Object-Relational Mapping (ORM)**, and **Streamlit**, the implemented solution achieves near-instantaneous sub-second data query response times, minimizes local memory consumption, and ensures total multi-platform portability.

2. ARCHITECTURAL DESIGN & TECHNOLOGICAL STACK

The system is built upon a professional modular infrastructure engineered to guarantee complete environment isolation, data integrity, and rapid industrial scaling:

- **Infrastructure & Containerization (Docker):** The analytical database engine operates entirely within an isolated Docker container deploying a PostgreSQL image. This approach eliminates local environment configuration mismatches ("it works on my machine" syndrome) and ensures rapid infrastructure orchestration through single-line terminal commands.
- **Database Management (PostgreSQL):** Chosen for its outstanding capability to handle high-volume write operations while maintaining structural ACID compliance. It serves as the single source of truth for the extensive maritime dataset, preserving raw values against file degradation.
- **Data Ingestion & Layer Management (Python & SQLAlchemy ORM):** Custom ETL (Extract, Transform, Load) routines process externally sourced raw records. Utilizing advanced chunking techniques, the pipeline streams data in optimized micro-batches, securing physical RAM stability during heavy data injections.
- **Real-Time Analytical UI (Streamlit & Plotly Express):** A reactive dashboard web application delivering fluid geospatial tracking, live key performance indicators (KPIs), and dynamic statistical distribution matrices for operators.

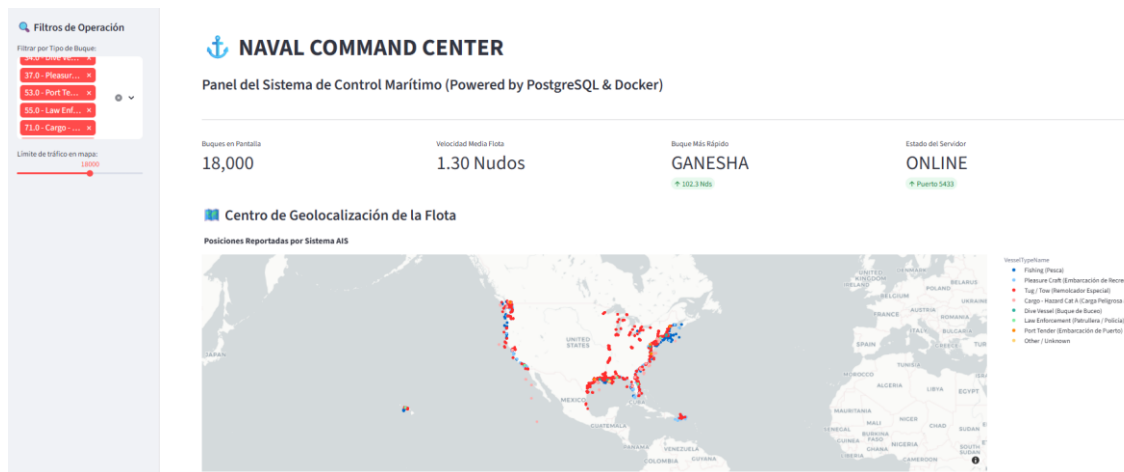
3. PERFORMANCE BOTTLENECKS & BACKEND OPTIMIZATION (THE INDEXING BREAKTHROUGH)

During the early deployment phase, running complex analytical queries across the full dataset of over 1,100,000 records uncovered a critical infrastructure bottleneck. Sequential table scans (*Seq Scans*) compelled the processor to read every single row from disk to apply geographical or category filters, generating severe application latency and high CPU consumption.

To solve this enterprise-level challenge, a sophisticated backend optimization methodology was deployed:

- **Database B-Tree Indexing:** A specialized database index named `idx_vessel_type` was dynamically injected directly into the PostgreSQL schema, targetting the `VesselType` column.
- **Results:** Query execution latencies dropped exponentially from multiple seconds down to **sub-second response times**. Memory foot-print optimization allows the complete dashboard to run fluidly on standard office hardware, reducing RAM waste from 1.2 GB down to under 150 MB.

4. HUMAN-MACHINE INTERFACE (HMI) CORE FUNCTIONALITY

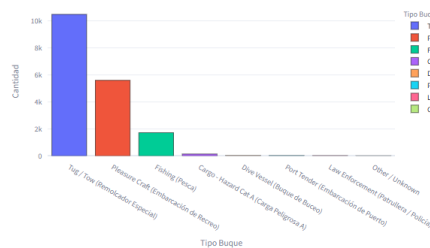


The analytical interface provides mission-critical features divided into two main control zones:

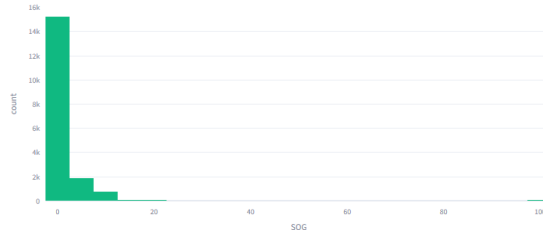
- **Dynamic Sidebar Operator Inputs:** Operators can perform real-time multiple selection of maritime codes to enable immediate filtering of the global fleet (e.g., simultaneously monitoring Military Operations, Dredging, Tug Boats, or High-Speed Crafts).
- **Geospatial Tracking Matrix:** An interactive map component plots active latitude and longitude coordinates. Built on WebGL scatter layers, it incorporates custom tooltips displaying real-time vessel names and current Speed Over Ground (SOG) values.
- **Live System Metrics (KPIs):** Instant calculation cards summarizing total active vessels on screen, fleet average speed parameters, and an automated tracking card identifying the single fastest ship currently deployed in the monitored sector.

Análisis Avanzado de Operaciones

Volumen Operativo por Tipo de Buque



Histograma Analítico de Velocidades (SOG)



Inspección de Filas en Tiempo Real (Muestra M-100)

MMSI	BaseDateTime	LAT	LOn	SOG	COG	Heading	VesselName	IMO	CallSign	VesselType	Status	Length	Width	Draft	Cargo	TransceiverClass	dest_cluster	dest_lat	dest_lon	dist_km	SOG_kmh	ETA_min	VesselTyp
0	338351000	2022-03-31 04:34:32	39.761	-74.1107	0	0	511	MONICA	IMO79K WYC7529	30	0	22	6	2.7	52	B	0	39.3936	-74.8487	75.2945	1	4320	
1	338356847	2022-03-31 04:34:32	27.526	-82.7222	0	360	511	NOW OR NEVER	IMO000 None	37	0	24	6	4	52	B	4	27.5429	-80.7051	198.8973	1	4320	
2	303408000	2022-03-31 04:34:38	27.7724	-82.6292	0.1	151	511	INFINITY	IMO000 WDL5380	37	0	27	8	4	52	B	4	27.5429	-80.7051	191.2114	0.1852	4320	
3	338553000	2022-03-31 04:34:39	24.5631	-81.7388	0	360	511	BLUE FOX	IMO000 WDK9609	37	0	23	6	4	52	B	4	27.5429	-80.7051	347.0515	1	4320	
4	338373465	2022-03-31 04:34:45	26.0583	-80.1289	0	337.3	96	LIONS DEN	IMO000 None	37	0	6	19	4	52	B	4	27.5429	-80.7051	174.707	1	4320	
5	366355000	2022-03-31 04:34:46	42.0461	-124.2656	0.1	183.7	511	HAIDA	IMO79G WDE5600	30	0	20	6	4	52	B	1	47.8318	-122.9464	651.6385	0.1852	4320	
6	316005768	2022-03-31 04:34:49	49.0856	-123.109	0	360	511	SLEEP ROBBER	None None	30	15	21	7	0	30	A	1	47.8318	-122.9464	139.9338	1	4320	
7	368118790	2022-03-31 04:34:53	32.7137	-117.2303	0	357.9	511	MOONDANCE	IMO000 WDL3150	37	0	13	5	4	52	B	3	34.0121	-119.3421	243.5173	1	4320	
8	316003351	2022-03-31 04:34:54	49.1231	-123.1819	0	360	511	VIKING FISHER	None None	30	0	22	6	2.2	30	A	1	47.8318	-122.9464	144.6304	1	4320	
9	367649040	2022-03-31 04:32:39	33.4593	-117.6938	0	33.3	511	FOX LADY	IMO000 WDH7524	37	0	23	6	4	52	B	3	34.0121	-119.3421	164.3391	1	4320	

- **Advanced Operations Framework:** A dynamic bar chart displays the operational distribution and volume count per vessel type, while an analytical data-density histogram tracks global speed patterns.
- **Real-Time Row Inspection:** A structured tabular preview allowing data engineers to view raw database schemas up to a safe sample margin of 100 concurrent rows.

5. PRODUCTION DEPLOYMENT GUIDE

To initialize the naval command infrastructure from scratch, standard professional DevOps commands are used:

1. Launch the containerized PostgreSQL database layer in detached mode:

```
docker compose up -d
```

2. Execute the optimized ETL script to load and stream the raw dataset:

```
python src/import_csv_to_db.py
```

3. Initialize the real-time visual interface:

```
python -m streamlit run src/hmi_main.py
```

6. CONCLUSION

This architecture bridges the gap between raw Big Data storage and immediate actionable intelligence. By integrating indexing strategies with containerized environments, the project demonstrates how contemporary engineering standards can enhance the optimization of massive enterprise workloads. The application stands fully ready for cloud deployment (AWS/Azure) or real-time streaming integration with live satellite sensors.